

Niranjan Madhavan Dindukurthu

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EXPERIENCE

Meta - Production Engineer

London, UK

Ads - Catalog

June '25 - present

- Prevented high-severity production incidents by building automated regression detection and validation gates that replay production traffic against code changes, blocking broken changes before they reach production.
- Reduced incident detection time by 40% and prevented \$5M in annual revenue loss by engineering automated rollout controls with real-time SLI monitoring and automated rollback on metric regression. Powers safe gradual deployments across every feature release across Meta.
- Saved \$3M annually in compute costs by building an AI-powered performance optimization system that leveraged profiled production workloads to identify bottlenecks and hot code paths, automatically generating optimized code changes across 1000+ services.
- Built end-to-end test automation frameworks integrated into release pipelines, increasing test coverage from <5% to 95% across critical ad-serving systems. Cut QA costs by ~\$150K/year while reducing incident detection time from weeks to minutes.
- Drove 25% reduction in mean-time-to-detection and 30% faster resolution by building unified observability (SLIs/SLOs, metrics, alerts, dashboards) and end-to-end distributed tracing, enabling faster root cause analysis across ad-serving services.
- Reduced operational costs and repeat incidents by converting on-call learnings into product and process improvements, driving root cause analysis on production outages, and sharing best practices across engineering teams.

Google - Systems Development Engineer

Bangalore, India

GCP - Google Distributed Cloud Hosted

May '24 - May '25

- Accelerated client deployments by 30% and reduced failures by 40% by engineering a metric-driven hardware validation system for Google's private cloud, integrated into the deployment pipeline.
- Designed and deployed a fleet-wide metrics collection pipeline using custom node exporters across diverse hardware classes, standardising ingestion into a unified observability platform and enabling automated alerting across the platform.
- Performed capacity planning and load analysis for GDCH deployments, modelling resource utilisation trends to right-size infrastructure and prevent capacity-related outages before customer impact.

Cisco - SRE Intern → SRE I → SRE II

India (Remote)

Network Engineering and Operations

Feb '21 - Mar '24

- Modernized core network infrastructure by migrating legacy IPAM systems to a scalable platform with an event-driven DNS backend processing millions of events per day, improving scalability and reliability for Cisco's global network.
- Managed on-prem Kubernetes clusters across multiple regions running production network services, maintaining cluster health and resolving pod/node-level failures to ensure high service availability.
- Eliminated cascading failures and improved reliability of the DNS service handling 2M QPS, by implementing client attributions and rate limiting, reducing service disruptions under high load.
- Cut MTTR by 60% by automating incident resolution workflows and building operational runbooks that enabled junior engineers to resolve incidents independently.

Manipal Institute of Technology - Research Assistant

Manipal, India

Software Defined Networks (SDN) - Security

Jan '20 - Jul '21

- Built a statistical intrusion detection system for SDN environments using DPDK-enabled Open vSwitch, detecting DDoS attacks 35% faster than traditional detection systems.

AWARDS AND RECOGNITION

- Received Amaze 2 award for 'Driving the highest level of achievement' for driving the IPAM integration project at Cisco.

SKILLS

- Languages:** Python, Go, Bash, SQL, Typescript.
- Systems:** Distributed systems design, fault-tolerant architectures, capacity planning, SLI/SLO frameworks.
- Infra/Platform:** Linux/Unix, Kubernetes, Docker, GCP, Terraform, Ansible, Helm, Jenkins, CI/CD, Git.
- Observability:** Prometheus, Grafana, Splunk, distributed tracing, alerting, SLO-based monitoring.

CERTIFICATIONS

- Cisco Certified Devnet Associate

EDUCATION

Manipal Institute of Technology, MAHE

Manipal, India

- Bachelor of Technology in Computer and Communication Engineering Aug '21
- Coursework: Distributed Systems, Data Structures and Algorithms, Computer Networks, Operating Systems, Database Management.